Arush Singh

SMART CONTRACT DEVELOPER

SOLIDITY & DEFI

github.com/04arush · linkedin.com/in/04arush
x.com/04arush · 04arush.github.io
04arush@gmail.com · New Delhi, India

PROFILE

Blockchain developer specialising in **Solidity smart contract** development, **DeFi protocols** and on-chain automation. Designed an intent-based batch auction exchange with **EIP-712** settlement, built a ZK-gated lending pool, and shipped a **Chainlink VRF & Automation** raffle. Grounded in contract security and rigorous **unit, fuzz and invariant testing** with Foundry.

SKILLS

Smart Contracts & Blockchain

Solidity · Foundry · DeFi · OpenZeppelin · Chainlink VRF / Automation / Oracle · Viem / Wagmi

Security & Testing

Slither · Unit / Fuzz / Invariant Testing · Echidna · CodeHawks · Ethernaut · CI Workflows

Protocol, ZK & Cryptography

Rust (arkworks) · Go (Cosmos SDK) · Noir (bb) · Circom · Groth16 / SNARKs · STARKs · PLONK · Poseidon · Merkle Sum Trees · Tokio · Alloy

Languages & Tooling

TypeScript · Node.js · React · Linux / Shell · Git · SQL (PostgreSQL) · NoSQL (MongoDB) · Python

EXPERIENCE

Blockchain Protocol Engineer

Oct 2026 - Present

Cubane · Hybrid · Part-time

[Tech used: Rust · Go · Solidity · Noir · TypeScript]

- Own critical components of Cubane's core architecture, working directly with the founders and the product and design teams.
- Building Cubane's Kosh wallet, and its block explorer, from protocol-level integrations through to the user-facing product.
- Work across the zero-knowledge and consensus layers of the protocol, alongside the Rust and Go systems around them.

PROJECTS

Intent-Based Batch Auction Exchange

↗ GitHub

Solidity · Foundry · OpenZeppelin · Slither · CI

- On-chain spot exchange where traders sign **EIP-712** orders off-chain and an allowlisted solver submits them per **epoch**; the contract settles every trade **atomically at one uniform clearing price** that it computes itself.
- Solver supplies only sorted, signed orders. The contract picks the price, allocates fills by priority, and rounds buyers up and sellers down, sending dust to the treasury.
- Internal-escrow design: settlement makes **no token transfers or untrusted calls**, removing reentrancy risk. Owner has no function to move or freeze balances.
- Defence in depth with limit and conservation checks, nonce burning, ERC-1271 wallet support, and fee-on-transfer rejection.

ZK Undercollateralized Lending Protocol

↗ GitHub

Noir · Solidity · Foundry · OpenZeppelin · Next.js

- Undercollateralized lending (**50% collateral** vs the usual 150%): `LendingPool.borrow()` verifies a Noir credit proof through a Barretenberg verifier before checking collateral and moving funds.
- Pool implements `deposit`, `borrow`, `repay` and `liquidate` with `ReentrancyGuard`, `SafeERC20`, strict naming conventions and full NatSpec.
- **100% line, branch and function coverage** across 23 tests: unit, fuzz, and invariant (128,000 actions per invariant) enforcing solvency, accounting integrity and collateral floors.

Provably Fair Raffle

↗ GitHub

Solidity · Foundry · Chainlink VRF v2.5 · Chainlink Automation

- Autonomous raffle with no admin intervention: **Chainlink Automation** triggers upkeep after each interval and **Chainlink VRF v2.5** selects a provably random winner, who receives the full pool before the raffle resets.
- Chain-aware deploy scripts via `HelperConfig` that also create, fund and register the VRF subscription.
- unit plus fuzz testing of state transitions and access control; deployed to Sepolia.

HACKATHONS

ETHOnline 2026

Participation Award

- Built **StreamProof**, a private royalty-tier verification protocol: a Noir circuit (Poseidon2 Merkle inclusion + threshold proof) verified on-chain via UltraHonk on Arc Testnet.

Frostbyte Hackathon

Finalist · Participation Award

- Qualified with the **Employee Payroll Manager** (Chainlink Automation), then advanced to the finale with the **ZK lending protocol**.

EDUCATION

Bachelor's — Computer Application

Aug 2023 - Ongoing

Indira Gandhi National Open University (IGNOU)
New Delhi, Delhi, IN

CERTIFICATIONS

Noir Programming & ZK Circuits	Cyfrin Updraft · Apr 2026
Fundamentals of Zero-Knowledge Proofs	Cyfrin Updraft · Mar 2026
Advanced Foundry	Cyfrin Updraft · Mar 2026
Foundry Fundamentals	Cyfrin Updraft · Mar 2026
Rust Programming Basics	Cyfrin Updraft · Feb 2026
Solidity Smart Contract Development	Cyfrin Updraft · Jan 2026
Blockchain Basics	Cyfrin Updraft · Dec 2025